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Graded Project – Part 1

AI Project Functional Methodologies

Framing of the Customer Churn Prediction project at RetailGenius

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Executive summary

RetailGenius sells a huge range of products around the world and collects a lot of data along the way: clicks, orders, support tickets, campaign results. The problem is that this data sits in different systems and its quality varies. The company picked churn prediction as the first project of its Data & AI program, and we think that is a good choice. The value is easy to measure and most of the data already exists.

Here is what we propose, in short.

- **Goal.** Bring the monthly churn rate of active customers down by about 15% (relative) within a year of going live, for example from 16.8% to around 14.3%, without blowing the retention budget.
- **Model.** A classifier that scores every active customer once a day. We will start with logistic regression as a baseline and expect a tree ensemble (random forest or gradient boosting) to win. Each score comes with the top reasons behind it, computed with SHAP, so marketing and support know what to do.
- **Way of working.** Scrum sprints of two weeks, organised around the CRISP-DM phases, with a Trello Kanban board. MLflow is used from the first experiment.
- **Roll-out.** Daily batch scores pushed to the CRM first, since it is cheap and safe. A real-time API for the support console comes later. Every release is tested against a control group before we claim any impact.
- **Governance.** A monthly steering committee, sprint reviews open to business users, a GDPR impact assessment, and a model card for every model in production.

1. Context and problem statement

1.1 What we learned from the interviews and workshops

Source	Key insight	Implication for the churn project
Business stakeholders	Decisions rely on demographics, purchase history, product performance and campaign effectiveness. KPIs: conversion, retention, revenue growth, CSAT.	Customer retention is already a board-level KPI → the project must report in business terms (retained revenue), not only model metrics.
Data team	Goals: personalisation, recommendation, fraud, inventory. Pain points: silos, data quality, privacy & security. A governance framework exists.	Data preparation will be the biggest workload; data owners and the governance framework must be involved early.
Technology team	SQL + NoSQL, cloud storage, data lake; ETL with Apache NiFi and Talend; scaling concerns.	Reuse existing ETL and data lake; the model must integrate into the existing stack and scale.
Product managers	Need real-time access to user behaviour data.	A real-time scoring option should be planned (phase 2).
Marketing executives	Need demographics and campaign engagement to target campaigns.	Main consumer of churn scores → retention campaigns, segment-level explanations.
Data consumers (support, finance, ops)	Each department has specific access needs.	Scores must be exposed per role (CRM, support console, BI), with access control.
Documentation analysis	Data dictionaries, data quality assessment, data flow diagrams and metadata catalog exist.	Speeds up data understanding; lineage helps for audit and GDPR.

1.2 Problem statement and churn definition

Customers at RetailGenius do not cancel a contract. They just stop buying, often without any clear signal. So before anything else we have to agree on what "churn" means, because that definition becomes the target the model learns from.

Element	Proposed definition (to validate with business)
Population	Active customers: at least one order in the last 12 months, account not closed.
Churn event	No order **and** no session in the next 90 days (window to be tuned with the purchase-frequency distribution; frequent buyers could use 60 days).
Prediction horizon	At date T, predict churn over T+1 → T+90 days, using data up to T only (no leakage).
Output	A churn probability (0–1), a risk segment (High / Medium / Low), and the top 3 reasons (SHAP).
Scoring frequency	Daily batch for all active customers; real-time on key events in phase 2.

2. Project strategy

2.1 Strategic objectives

We tried to make each objective measurable and tie it back to what the business told us matters (retention and growth).

#	Strategic objective	Measurable target (indicative)
O1	Reduce customer churn among active customers	−15% relative churn rate within 12 months of go-live, measured against a control group
O2	Protect revenue and customer lifetime value (CLV)	Retained revenue ≥ 3× the cost of retention actions (ROI ≥ 3)
O3	Make retention proactive and targeted instead of mass campaigns	≥ 60% of retention budget spent on customers flagged by the model; −30% cost per retained customer
O4	Understand why customers leave	Top churn drivers published monthly to product, marketing and support
O5	Build a reusable Data & AI foundation (first project of the program)	Customer 360 dataset, feature store and MLOps pipeline reusable for recommendation and fraud projects
O6	Stay compliant and trustworthy	100% GDPR compliance, explainable predictions, zero use of sensitive attributes

2.2 Key performance indicators

We track KPIs on three levels. A model can look great on paper and still change nothing if nobody uses its scores, so model metrics alone are not enough.

Model (technical) KPIs

KPI	Why it matters	Target
Recall (churners detected)	Missing a churner = lost customer. Recall is the priority metric.	≥ 0.75
Precision	False alarms waste retention budget (discounts given to loyal customers).	≥ 0.50 at chosen threshold
F1-score / PR-AUC	Single summary for imbalanced classes (~17% churners in the reference dataset).	PR-AUC ≥ 0.70
ROC-AUC	Ranking quality, threshold independent.	≥ 0.85
Lift @ top 10%	How much better than random targeting the top-risk decile is.	≥ 3×

KPI	Why it matters	Target
Calibration (Brier score)	Probabilities must be reliable to compute expected value of an action.	Brier ≤ 0.10

Business KPIs

KPI	Definition
Churn rate	Share of active customers who churn in the 90-day window (treated vs control group).
Retention rate / save rate	Share of contacted high-risk customers who order again within 90 days.
Retained revenue & CLV	Revenue from saved customers; CLV uplift for treated group.
Retention ROI	(Retained margin − campaign cost) / campaign cost.
Customer satisfaction (CSAT / NPS)	Does targeted retention improve satisfaction of at-risk customers?
Repeat purchase rate, conversion rate	Leading indicators already followed by the business.

Adoption and operational KPIs

- Share of high-risk customers actually contacted by marketing / support (target ≥ 80%).
- Pipeline reliability: daily scoring completed on time (SLA ≥ 99%); API latency < 200 ms (phase 2).
- Data freshness and data quality score of the input features.
- User satisfaction of marketing and support teams with the tool (quarterly survey).

2.3 How AI contributes to customer retention

AI contribution	Concrete use at RetailGenius
Early warning	Detect weak signals (fewer sessions, longer time since last order, complaints, cart abandonment) weeks before a customer leaves, impossible to follow by hand for millions of customers.
Prioritisation	Rank customers by churn risk × customer value so that limited budget goes to customers who are both at risk and valuable.
Personalised actions	Use SHAP reasons to choose the right action: delivery issue → support call; price sensitivity → coupon; low engagement → personalised recommendations.
Root-cause insights	Global explanations reveal structural drivers (e.g. complaints, delivery distance, payment method) that product and operations teams can fix.
Measurement and learning	A/B tests and, later, uplift modelling identify who actually responds to an action, avoiding discounts to customers who would stay anyway.
Better customer experience	Combined with recommendation and support data, retention becomes part of a personalised journey instead of generic e-mails.

3. Project design

Figure 1 shows the architecture we are aiming for. It reuses the data lake and ETL tools RetailGenius already has rather than building a parallel stack.

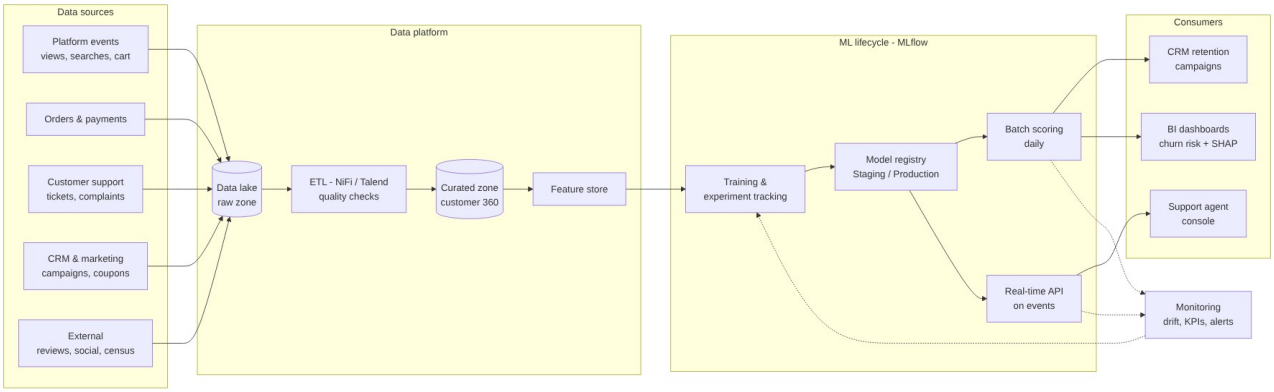


Figure 1 – Target architecture of the churn prediction solution

3.1 Data

Relevant data sources

Domain	Source / system	Example features	Priority
Customer profile	CRM, account database (SQL)	Tenure, city tier, preferred device, preferred payment mode, number of addresses, marital status	High
Transactions	Order management, payments	Recency (days since last order), frequency, monetary value, order count growth, coupons used, cashback, preferred category	High
Behaviour on platform	Clickstream / events (NoSQL, data lake)	Sessions & hours on app, product views, searches, cart abandonment, devices registered	High
Customer support	Ticketing system	Complaints (yes/no, count), time to resolution, contact reason, satisfaction after ticket	High
Satisfaction & reviews	Survey tool, internal reviews	Satisfaction score, review ratings, sentiment of review text	Medium
Marketing	Campaign tools, e-mail platform	Campaign exposure, open / click rates, unsubscriptions, offers accepted	Medium
Logistics	Order processing, sellers	Warehouse-to-home distance, delivery delays, returns, seller rating	Medium
External	Review platforms, social media APIs, census, market data	External sentiment, regional demographics, competitor price index, seasonality	Low (phase 2)

The Kaggle E-Commerce Customer Churn dataset we use in Part 2 covers part of this list: tenure, city tier, distance to the warehouse, time on the app, satisfaction, complaints, days since last order, cashback and a few others. It is a fair stand-in for the curated customer table we would build at RetailGenius.

Challenges in using the data

Challenge	Description	Mitigation
Data silos	Clickstream, CRM, support and logistics data live in different systems and teams.	Build a single customer 360 table in the curated zone of the data lake via existing NiFi/Talend pipelines; one unique customer ID.

Challenge	Description	Mitigation
Data quality	Missing values, inconsistencies (e.g. "Phone" vs "Mobile Phone"), outliers, duplicated customers.	Automated quality checks (Great Expectations-type rules), imputation strategy, harmonised categories, data quality KPIs per source.
Defining the label	No explicit cancellation in e-commerce; the churn window is a business choice.	Analyse purchase-frequency distribution, validate definition with marketing, test sensitivity of results to window size.
Class imbalance	Churners are a minority (~17%).	Stratified splits, class weights or SMOTE, metrics like recall / PR-AUC instead of accuracy.
Data leakage	Features computed after the prediction date (e.g. "account closed") make the model look perfect offline.	Point-in-time feature computation, strict time-based train/test split, feature review with domain experts.
Privacy & regulation	Personal data under GDPR (and local laws in other countries); profiling rules (art. 22).	Legal basis (legitimate interest), DPIA, pseudonymisation, data minimisation, no sensitive attributes, retention periods, opt-out.
Security & access	Many consumers; risk of leaking customer data.	Role-based access, encryption at rest and in transit, audit logs (already in place, extend to the ML pipeline).
Volume & latency	Huge event volumes; real-time needs from product teams.	Aggregate events into daily features; distributed processing (Spark) and a feature store for online/offline consistency.
External data	Licensing, terms of service of social APIs, scraping legality, matching with internal customers.	Use aggregated external data at region/category level; legal review before any collection.

3.2 Models

Suitable AI models

This is a supervised binary classification problem on tabular data: each customer either churns (1) or stays (0). We will compare a simple model that is easy to explain with stronger ensemble models, and pick based on both performance and how well we can explain the results.

Model	Strengths	Limits	Role in project
Logistic regression	Simple, fast, coefficients easy to explain, good calibration	Linear; needs feature engineering for interactions	Baseline
Decision tree	Very intuitive rules for business	Unstable, overfits	Communication / rule extraction
Random Forest	Robust, handles non-linearities and mixed features, little tuning	Larger models, less calibrated	Strong candidate
Gradient boosting (XGBoost / LightGBM / CatBoost)	State of the art on tabular data, handles missing values, fast inference, works with SHAP TreeExplainer	More hyper-parameters, risk of overfitting	**Target model**
Neural networks / sequence models (LSTM, transformers on event sequences)	Can use raw clickstream sequences	Data-hungry, costly, hard to explain	Later exploration
Survival analysis (Cox, survival forests)	Predicts **when** a customer will churn	Less familiar to business users	Phase 2 option

Model	Strengths	Limits	Role in project
Uplift models	Predicts who will respond to a retention action	Requires A/B test data	Phase 2 after first campaigns

Training, validation and testing

1. **Data split:** a time-based split mirrors production: train on older snapshots, validate on the next period, test on the most recent period never seen during development (e.g. 70 / 15 / 15). For the static public dataset we use a stratified random split.
2. **Pre-processing pipeline:** imputation, encoding of categorical variables and scaling are fitted on the training set only, inside a scikit-learn Pipeline, so the exact same steps are applied at inference.
3. **Cross-validation & tuning:** stratified k-fold (k = 5) on the training set, with randomized or Bayesian search (Optuna) for hyper-parameters. Every run is logged in MLflow (parameters, metrics, artifacts).
4. **Imbalance handling:** class weights or SMOTE (on training folds only), then choose the decision threshold that maximises expected business value (cost of a missed churner vs cost of an unnecessary offer).
5. **Evaluation:** recall, precision, F1, ROC-AUC, PR-AUC, lift and calibration on the hold-out test set; error analysis by segment (country, city tier, device) to check fairness and stability.
6. **Business validation:** shadow scoring, then an A/B test on real customers (treated vs control) before full roll-out This is the only way to prove the model actually reduces churn.

Model versioning and serving

Topic	Practice
Code versioning	Git + GitHub, feature branches, pull requests with code review, CI (lint, tests).
Data versioning	Dated snapshots in the data lake (or DVC) and the dataset hash logged with each MLflow run, so any model can be reproduced.
Experiment tracking	MLflow Tracking: parameters, metrics, plots, environment (conda / poetry lock file).
Model versioning	MLflow Model Registry: each trained model gets a version; aliases / stages None → Staging → Production → Archived; approval by the model owner before promotion.
Documentation	A model card per production version: training data, metrics, limits, fairness checks, owner.
Serving	Batch: a scheduled job (Airflow) loads the "Production" model from the registry and scores all customers daily. Real-time: the same model packaged as a REST API (MLflow serving / FastAPI in Docker, on Kubernetes or a managed cloud endpoint).
Rollback	Previous production version kept in the registry; switching the alias back is a one-step rollback.

3.3 Deployment

Deployment strategies

Strategy	Description	Pros	Cons / when to use
Batch scoring	Daily job scores all active customers; results written to CRM and BI.	Simple, cheap, fits weekly campaigns	Not reactive within the day → **phase 1 choice**
Real-time (online) API	Score computed on events (login, cart abandon, complaint).	Immediate actions in support console / app	Needs online feature store, low latency → **phase 2**
Embedded / streaming	Model running inside a stream (Kafka) pipeline.	Very low latency at scale	Complex; only if real-time volume justifies it
Shadow deployment	New model scores in parallel without being used.	Safe comparison with current model	Used before each first release

Strategy	Description	Pros	Cons / when to use
Canary release	New model serves a small % of traffic, then increases.	Limits blast radius	For real-time API updates
Blue / green	Two identical environments, instant switch.	Zero downtime, quick rollback	Double infrastructure cost
A/B test (champion / challenger)	Customers randomly split between models or between action vs no action.	Measures true business impact	Requires time and enough volume

Production considerations

- **Integration:** outputs written into existing systems (CRM, support console, BI) so that users do not need a new tool which is what the technology team asked for.
- **Scalability & performance:** millions of customers scored daily; containerised services, autoscaling, distributed processing for feature computation.
- **Reliability:** orchestration with retries and alerts, SLAs, fallback rules (e.g. previous day's score) if the pipeline fails.
- **Training/serving consistency:** the same feature code and pre-processing pipeline for training and inference (feature store, packaged pipeline).
- **Security:** authentication on the API, secrets management, encryption, least-privilege access to customer data.
- **Privacy & ethics:** GDPR (purpose limitation, right to object to profiling), no discriminatory features, human-in-the-loop for decisions with significant impact.
- **Explainability:** SHAP reasons delivered with each score so marketing and support can act and justify actions.
- **Cost:** cloud compute and storage budget, cost of retention actions (discounts). Both go into the ROI.
- **CI/CD for ML:** automated tests (unit, data validation, model performance gate) before any promotion to production.
- **Change management:** training of marketing and support users, retention playbooks, feedback loop on actions taken.

3.4 Monitoring

Monitoring performance over time

Layer	What we monitor	Tool / frequency
Data quality	Missing values, schema changes, out-of-range values, volumes per source	Automated checks on each pipeline run
Data drift	Distribution change of input features (PSI, Kolmogorov–Smirnov, Jensen–Shannon)	Daily, dashboard (Evidently AI / custom)
Prediction drift	Distribution of scores and share of high-risk customers	Daily
Model performance	Recall, precision, PR-AUC, calibration once the 90-day labels are known	Monthly (labels arrive with delay)
Business impact	Churn rate treated vs control, retention ROI, retained revenue	Monthly steering committee
Operations	Job success, latency, errors, infrastructure cost	Real time (alerts)
Fairness	Performance by segment (country, age group, device)	Quarterly

Plan for model drift and accuracy maintenance

1. **Detect:** thresholds and alerts, e.g. PSI > 0.2 on a top-10 feature, recall drop > 5 points vs. reference, or a sudden change in the share of high-risk customers.
2. **Diagnose:** the data scientist on duty checks whether the cause is a data problem (broken pipeline, new category), a real change in customer behaviour (new competitor, season, price change) or a feedback effect (retention actions changing the labels).
3. **Act:** fix the data if it is a pipeline issue; otherwise retrain on recent data, review features and threshold. Retraining is also scheduled every quarter even without alerts.
4. **Validate:** the retrained "challenger" model must beat the "champion" on a recent hold-out set and pass fairness and calibration checks; then shadow / canary release.
5. **Promote and document:** new version registered in MLflow, model card updated, stakeholders informed in the steering committee.
6. **Keep a control group:** a small random group of customers never receives model-driven actions, so that true churn and model impact can always be measured without bias.

4. Project team

4.1 Roles, expertise and skills

Role	Responsibilities	Key skills and expertise
Project sponsor (CMO / Chief Customer Officer)	Owns the business objective, budget and priorities; arbitrates	Business vision, decision making
Product owner (AI product manager)	Translates business needs into backlog, defines success KPIs, prioritises	Retail domain, agile, data literacy, communication
Project manager / Scrum master	Plans, removes blockers, runs ceremonies, tracks risks and costs	Agile & Kanban, risk management, facilitation
Data engineer(s)	Builds pipelines and customer 360 table, data quality, feature store	SQL / NoSQL, Spark, NiFi / Talend, cloud, data modelling
Data scientist(s)	Exploration, feature engineering, modelling, evaluation, explainability	Statistics, Python, scikit-learn / XGBoost, SHAP, experimentation
Data analyst	EDA, KPI dashboards, A/B test analysis, business reporting	SQL, BI tools (Power BI / Tableau), statistics, storytelling
ML engineer / MLOps	Packaging, CI/CD, serving, monitoring, MLflow	Docker, Kubernetes, MLflow, Airflow, cloud, software engineering
Software / CRM integration engineer	Integrates scores into CRM, support console and apps	APIs, CRM platforms, back-end development
Data protection officer & legal	GDPR compliance, DPIA, profiling rules	Privacy law, risk assessment
Security engineer	Access control, encryption, security reviews	Cloud security, IAM
Business experts (marketing, customer support)	Define churn and actions, validate features and results, run campaigns	Customer knowledge, campaign management
UX / change manager (part time)	Design how scores are shown, train users	UX, training, communication

4.2 Cross-functional collaboration

- **One cross-functional squad** with business experts embedded (a marketing analyst and a support lead part time), rather than a data team working in isolation.

- **Shared tools:** Trello board visible to everybody, shared documentation (Confluence / Google Docs), GitHub for code, MLflow UI for experiments.
- **Shared rituals:** daily stand-up (team), sprint review with business users every 2 weeks, retrospectives.
- **Clear responsibilities** with a RACI matrix (below) and a definition of done that includes documentation, tests and business validation.
- **Common vocabulary:** a short glossary (churn, recall, threshold, drift…) and data literacy sessions for business teams; domain sessions for the data team.

Activity	Sponsor	PO	Data Eng	Data Sci	MLOps	Marketing / Support	DPO
Churn definition & KPIs	A	R	C	C	I	C	I
Data pipelines & quality	I	A	R	C	C	I	C
Model development	I	A	C	R	C	C	I
Deployment & monitoring	I	A	C	C	R	I	I
Retention campaigns	A	C	I	C	I	R	I
GDPR compliance	A	R	C	C	C	I	R

Table – RACI matrix (R = Responsible, A = Accountable, C = Consulted, I = Informed)

4.3 Ensuring team alignment with project strategy

- A **project charter** and a one-page **KPI tree** (company goal → retention → model KPIs) shared at kick-off.
- **Objectives and key results (OKRs)** per quarter, derived from strategic objectives O1–O6.
- **Backlog prioritised by business value** by the product owner; each user story is linked to an objective.
- **Sprint goals** reviewed against KPIs in sprint reviews; the steering committee re-validates priorities monthly.
- Transparent **dashboards** of progress and model / business KPIs.

4.4 Collaboration with other departments

Department	What they give to the AI team	What they receive	Collaboration mechanism
Marketing	Campaign history, retention offers, budget, feedback on campaign results	Daily risk scores, segments, reasons, A/B test results	Joint retention playbook; bi-weekly review; marketing analyst in squad
Customer support	Ticket data, qualitative insights on why customers leave	Risk flag and top reasons in the agent console	Pilot with a group of agents; feedback form in console
Product	Feature changes, release calendar	Churn drivers linked to product experience	Monthly insight session
Finance	Margin data, cost of actions	ROI of retention program	Quarterly ROI review
IT / technology	Infrastructure, integration, security	Requirements, deployment packages	Architecture board, shared DevOps tooling
Legal / DPO	Compliance guidance	DPIA inputs, model cards	Review at each major release

5. Project governance and communication

5.1 Key stakeholders

Stakeholder	Interest / expectation	Influence	Strategy
Executive committee / sponsor	Retention, revenue growth, ROI	High	Manage closely
Marketing executives	Targeted campaigns, segment insights	High	Manage closely
Customer support management	Prioritised list of at-risk customers	Medium	Keep satisfied & involved
Product managers	Real-time behaviour data, churn drivers	Medium	Keep informed
Data team	Data quality, governance, scalable models	High	Core team
Technology & infrastructure team	Integration, security, scalability	High	Manage closely
Finance	Costs and ROI	Medium	Keep satisfied
Legal / DPO / security	Compliance, risk	High (veto)	Keep satisfied, early involvement
End customers (indirect)	Relevant offers, privacy respect	Indirect	Transparency, opt-out

5.2 Governance instances

Instance	Participants	Frequency	Purpose / decisions
Steering committee	Sponsor, marketing & support heads, CTO, PO, project manager, DPO	Monthly	Validate priorities, budget, go / no-go for production, review business KPIs and risks
Project committee	PO, project manager, tech leads, business referents	Weekly	Progress, planning, risks, dependencies
Sprint review / demo	Squad + business users	Every 2 weeks	Demo of increments, feedback, backlog adjustment
Sprint planning & retrospective	Squad	Every 2 weeks	Commit to sprint goal, improve the way of working
Daily stand-up	Squad	Daily (15 min)	Coordination, blockers
Technical / architecture design authority	Tech leads, architects, security	On demand / monthly	Validate architecture, security and integration choices
Data & AI ethics / compliance review	DPO, legal, PO, data scientist	At each major release	DPIA, fairness, model card, approval for production
Model performance review (run phase)	MLOps, data scientists, PO	Monthly	Drift, performance, retraining decisions

5.3 Communication plan

Audience	Message / content	Channel	Frequency	Owner
Executive committee	Business KPIs, ROI, milestones, risks	Steering deck, one-page dashboard	Monthly	PO / sponsor

Audience	Message / content	Channel	Frequency	Owner
Marketing & support teams	Risk lists, playbooks, campaign results, tool training	CRM / console, workshops, newsletter	Daily data; bi-weekly reviews	PO, data analyst
Project team	Tasks, blockers, decisions	Trello, Slack / Teams, stand-up	Daily	Scrum master
IT & data teams	Architecture, releases, incidents	Technical docs, GitHub, release notes	Per release	Tech lead
Legal / DPO	Data used, purpose, risks	DPIA, model card	Per release	PO + data scientist
Whole company	Vision and success stories of the Data & AI program	Internal newsletter, town hall	Quarterly	Sponsor

5.4 Communicating model outputs

To technical teams

- MLflow UI with all runs, metrics (ROC/PR curves, confusion matrix, calibration plot), parameters and model versions.
- Technical documentation (Sphinx), model card, API specification, monitoring dashboards and alerts.
- Global and local SHAP plots (summary, beeswarm, dependence) to debug the model and detect leakage.

To non-technical teams

- **Simple outputs:** risk level (High / Medium / Low) instead of raw probability, plus the **top 3 reasons in plain language** (e.g. "No order for 25 days", "Complaint last month", "Low cashback").
- **Recommended action** attached to each segment via the retention playbook.
- **Business dashboards:** churn rate trend, number of at-risk customers, retained revenue, campaign results, with no technical jargon.
- **Storytelling with examples:** customer personas and before/after stories in steering meetings.
- **Honest communication of uncertainty:** "out of 10 customers flagged High, about 6 would actually leave". That explains precision without using the word.

6. AI project management methodology

6.1 Chosen methodology: Agile Scrum + CRISP-DM, with Kanban and MLOps

We went for a mix. CRISP-DM gives us the phases of a data project (business understanding, data understanding, preparation, modelling, evaluation, deployment). Scrum gives us the rhythm: two-week sprints, a product owner, a review with the business at the end of each sprint. A Trello Kanban board makes the work visible to everyone. Once the model is live, the run phase (monitoring, retraining) moves to a simple Kanban flow.

Methodology	Strengths	Weaknesses for this project
Waterfall	Clear plan and budget, good for fixed scope	AI results are uncertain; late feedback; not suited to experimentation
CRISP-DM alone	Standard for data mining, business-first	No team organisation or delivery rhythm; little on deployment and monitoring
Scrum alone	Short iterations, frequent feedback	Research tasks are hard to estimate; no data-specific phases
Kanban	Flexible flow, good visibility, WIP limits	Less structure for planning and deadlines

Methodology	Strengths	Weaknesses for this project
Hybrid Scrum + CRISP-DM + Kanban (chosen)	Structure of data phases + agility + visibility; fits MLOps loop	Requires discipline and a mature product owner

6.2 Why it is suitable

- We do not know upfront which data or model will work. Short sprints let us find out quickly and change direction if needed.
- Marketing and support see the work every two weeks in the sprint review. We need them for the churn definition and for deciding what to do with at-risk customers.
- Machine learning work loops back on itself: a bad evaluation usually sends you back to data preparation. CRISP-DM already expects this.
- A first usable version (daily scores from a simple model) can be ready in about three months and improved after that.
- The same board carries on after go-live for monitoring and retraining, so nothing gets lost in a handover.

6.3 Indicative roadmap

Phase (CRISP-DM)	Sprints	Main deliverables
0. Framing / business understanding	S0 (2 weeks)	Charter, churn definition, KPIs, stakeholder map, DPIA started
1. Data understanding	S1–S2	Data inventory, access, EDA, data quality report
2. Data preparation	S2–S4	Customer 360 table, feature pipeline, label generation
3. Modelling	S3–S5	Baselines, tuned model, MLflow experiments, SHAP analysis
4. Evaluation	S5–S6	Test-set results, business validation, model card, go / no-go
5. Deployment (MVP)	S6–S7	Daily batch scoring to CRM, dashboards, pilot campaign with A/B test
6. Monitoring & improvement	Continuous (Kanban)	Drift monitoring, retraining, real-time API, uplift model

6.4 Risks and mitigation strategies

Risk	Prob.	Impact	Mitigation
Poor data quality / silos delay the project	High	High	Data audit in sprint 1, start with available core sources, data owners committed in charter
Ambiguous churn definition	Medium	High	Validate definition with business in sprint 0; sensitivity analysis
Data leakage → over-optimistic results	Medium	High	Point-in-time features, time-based split, peer review
Model not good enough to be useful	Medium	High	Clear success thresholds, baselines, early go / no-go; value even from simple rules
Low adoption by marketing / support	Medium	High	Co-design, explainable outputs, training, integrate into existing tools
Retention actions not effective (or give discounts to loyal customers)	Medium	Medium	A/B tests, control group, later uplift modelling
GDPR / privacy non-compliance	Low	Very high	DPO involved early, DPIA, data minimisation, opt-out

Risk	Prob.	Impact	Mitigation
Bias / unfair treatment of customer groups	Medium	High	Exclude sensitive attributes, fairness checks by segment
Model drift after go-live	High	Medium	Monitoring, alerts, scheduled retraining
Integration problems with legacy systems	Medium	Medium	Involve IT from sprint 0, batch first, API contracts
Budget / planning overrun due to iterations	Medium	Medium	Time-boxing, MVP scope, contingency (see 6.5)
Key-person dependency	Medium	Medium	Documentation, code reviews, pair work, shared repo

6.5 Handling costs and planning drift due to model iterations

- **Time-boxing experiments:** each modelling spike has a fixed duration (e.g. one sprint) and a success criterion defined in advance; if not reached, we stop or pivot.
- **MVP first:** deliver a simple but useful model early; improvements are prioritised in the backlog by expected value.
- **Contingency:** keep a 15–20% budget and time buffer for exploration and re-work, visible in the steering committee.
- **Stage gates / go-no-go:** at the end of evaluation, the steering committee decides to continue, re-scope or stop, based on KPIs.
- **Track costs per iteration:** team days, cloud compute (track training costs in MLflow tags), and cost of retention actions; compare with expected retained revenue.
- **Re-planning with velocity:** update forecasts each sprint (burn-up chart) and communicate changes early.
- **Reuse:** feature store, pipelines and templates (cookiecutter) cut the cost of each new iteration and of future AI projects.

7. Mock project on Trello (Kanban board)

To simulate the project we built a Kanban board covering everything from framing to monitoring. It has six lists, a colour label per phase and an owner on each card. Figure 2 shows the board around sprint 4, while modelling is under way.

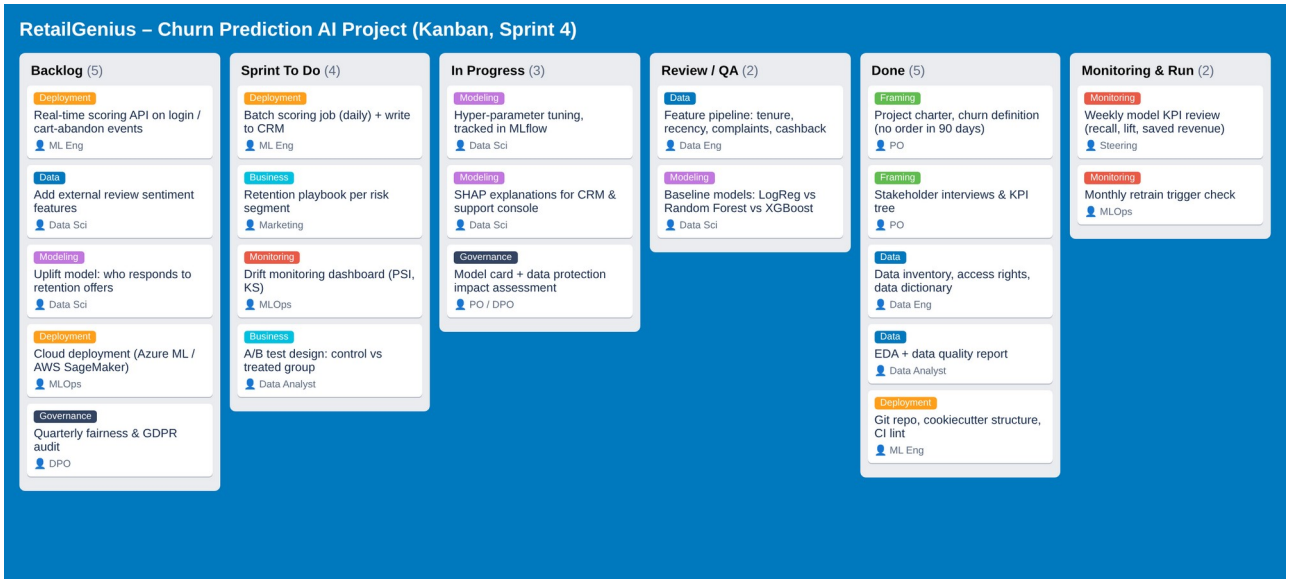


Figure 2 – Kanban board for the churn prediction project (sprint 4 snapshot)

Trello board link: _____ (add the group's public board URL and a screenshot here).

Board set-up

Element	Configuration
Lists	Backlog → Sprint To Do → In Progress → Review / QA → Done → Monitoring & Run
Labels	Framing (green), Data (blue), Modeling (purple), Deployment (orange), Monitoring (red), Governance (dark), Business (cyan)
Card content	User story, acceptance criteria / definition of done (checklist), owner, due date, link to GitHub / MLflow
WIP limit	Max 3 cards in "In Progress" and 3 in "Review / QA" to keep the flow
Definition of done	Code reviewed and merged, tests pass, documentation updated, run logged in MLflow, validated by PO
Power-ups (optional)	Calendar, GitHub integration, custom fields (story points, sprint)

Full card list

List	Card	Label	Owner
Backlog	Real-time scoring API on login / cart-abandon events	Deployment	ML Eng
Backlog	Add external review sentiment features	Data	Data Sci
Backlog	Uplift model: who responds to retention offers	Modeling	Data Sci
Backlog	Cloud deployment (Azure ML / AWS SageMaker)	Deployment	MLOps
Backlog	Quarterly fairness & GDPR audit	Governance	DPO
Sprint To Do	Batch scoring job (daily) + write to CRM	Deployment	ML Eng

List	Card	Label	Owner
Sprint To Do	Retention playbook per risk segment	Business	Marketing
Sprint To Do	Drift monitoring dashboard (PSI, KS)	Monitoring	MLOps
Sprint To Do	A/B test design: control vs treated group	Business	Data Analyst
In Progress	Hyper-parameter tuning, tracked in MLflow	Modeling	Data Sci
In Progress	SHAP explanations for CRM & support console	Modeling	Data Sci
In Progress	Model card + data protection impact assessment	Governance	PO / DPO
Review / QA	Feature pipeline: tenure, recency, complaints, cashback	Data	Data Eng
Review / QA	Baseline models: LogReg vs Random Forest vs XGBoost	Modeling	Data Sci
Done	Project charter, churn definition (no order in 90 days)	Framing	PO
Done	Stakeholder interviews & KPI tree	Framing	PO
Done	Data inventory, access rights, data dictionary	Data	Data Eng
Done	EDA + data quality report	Data	Data Analyst
Done	Git repo, cookiecutter structure, CI lint	Deployment	ML Eng
Monitoring & Run	Weekly model KPI review (recall, lift, saved revenue)	Monitoring	Steering
Monitoring & Run	Monthly retrain trigger check	Monitoring	MLOps

Example card detail – "Baseline models: LogReg vs Random Forest vs XGBoost"

- **User story:** As a marketing manager, I want a first churn model with known performance so that I can judge whether targeted retention is possible.
- **Checklist:** stratified split done · LogReg baseline logged in MLflow · RF and XGBoost logged · metrics compared (recall, PR-AUC) · results presented in sprint review.
- **Acceptance criteria:** recall ≥ 0.75 and PR-AUC ≥ 0.70 on the validation set; no leakage feature.

8. Conclusion

We think churn is a good first AI project for RetailGenius. The payoff is direct and measurable, the data is already there, and the pieces it forces the company to build (a customer 360 table, a feature store, an MLOps pipeline, governance rules) will be reused by the next projects. In our view, the algorithm is not where this project will succeed or fail. What matters more is a clear churn definition, clean data, real involvement from marketing and support, and an honest measurement of impact with a control group. Parts 2 and 3 cover the technical side: a churn model tracked with MLflow, then explained with SHAP.

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[11] Generative AI (Claude, Anthropic) was used to help structure and draft parts of this report. The group reviewed and edited all content.